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| **Grade** | **15.00** out of 15.00 (**100**%) |

Question **1**

Correct

Mark 1.00 out of 1.00

In binary search, how is the middle element determined?

a. By using a hash function
b. By comparing each element sequentially
c. By dividing the list length by two ✓
d. By starting from the first element

Question **2**

Correct

Mark 1.00 out of 1.00

Which of the following is not a limitation of binary search algorithm?

a. There must be a mechanism to access middle element directly
b. Binary search algorithm is not efficient when the data elements more than 1500 ✓
c. Must use a sorted array
d. Requirement of sorted array is expensive when a lot of insertion and deletions are needed

Question **3**

Correct

Mark 1.00 out of 1.00

In linear search, if the target element is not found in the list, what is the result?

a. The last element is returned
b. An error is raised
c. The first element is returned
d. The search is considered unsuccessful ✓

Question **4**

Correct

Mark 1.00 out of 1.00

In binary search, what happens if the middle element does not match the target element?

- a. The search continues in the left or right sublist ✓
- b. The list is sorted
- c. The search stops
- d. The search continues from the beginning

Question **5**

Correct

Mark 1.00 out of 1.00

What happens when the element is found in linear search?

- a. The search backtracks to find duplicate elements
- b. The search starts over from the beginning
- c. The search continues until the end of the list
- d. The search stops immediately ✓

Question **6**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

- a. It is slower than linear search for small lists
- b. It requires the list to be sorted ✓
- c. It does not work with negative numbers
- d. It can only be applied to large lists

Question **7**

Correct

Mark 1.00 out of 1.00

Which of the following is a conventional searching technique?

- a. Binary search
- b. Hashing
- c. Dynamic search
- d. Linear search ✓

Question **8**

Correct

Mark 1.00 out of 1.00

What is searching in the context of computer science?

a. Deleting elements from a list
b. Determining whether an element is present in a list ✓
c. Sorting elements in a list
d. Inserting elements into a list

Question **9**

Correct

Mark 1.00 out of 1.00

In which situation is linear search more efficient than binary search?

a. When the list is large and sorted
b. When the list is small and sorted
c. When the list is large and unsorted
d. When the list is small and unsorted ✓

Question **10**

Correct

Mark 1.00 out of 1.00

What is the key characteristic of binary search?

a. It compares elements sequentially
b. It works on unsorted lists
c. It can be applied only if the list is sorted ✓
d. It always starts from the beginning of the list

Question **11**

Correct

Mark 1.00 out of 1.00

What is the advantage of binary search over linear search?

a. Binary search does not require dividing the list
b. Binary search works on unsorted lists
c. Binary search can find multiple instances of the target element
d. Binary search has a lower time complexity on large, sorted lists ✓

Question **12**

Correct

Mark 1.00 out of 1.00

Finding the location of a given item in a collection of items is called

a. Mining
b. Searching ✓
c. Finding
d. Discovering

Question **13**

Correct

Mark 1.00 out of 1.00

Which method of searching involves sequentially comparing each element until a match is found?

a. Linear search ✓
b. Binary search
c. Jump search
d. Hashing

Question **14**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

a. Linear search requires the list to be sorted.
b. Linear search divides the list into halves.
c. Linear search can be applied to both sorted and unsorted lists. ✓
d. Linear search is more efficient than binary search on large lists.

Question **15**

Correct

Mark 1.00 out of 1.00

If a list contains 1000 elements, how many comparisons would a binary search typically make in the worst case?

a. 100
b. 1000
c. 10 ✓
d. 500